1 DH description

The DH frames are assigned from the CuRo6R URDF, with frame $\{0\}$ at `base_link`, joint i rotating about z_{i-1}, and $q = 0$ matching the URDF home configuration. We use standard DH transforms:

$$^{i-1}T_i = R_z(\theta_i)\, T_z(d_i)\, T_x(a_i)\, R_x(\alpha_i). \tag{1}$$

Table 1 gives the resulting parameters and the geometric choices that determine them. The only fixed joint-angle offset is $\theta_3 - q_3 = \pi/2$. The full sequence of frame drawings is given in Appendix A.

Table 1: DH parameters of CuRo6R. Lengths are in metres; angular constants are in radians.

i	d_i	θ_i	α_i	a_i	Geometric origin of the parameters
1	1	q_1	$-\pi/3$	0.8	The two base heights add: $d_1 = 0.5 + 0.5$; the lateral offset is $a_1 = 0.8$. Joint 2's axis is its local y axis, so the 30° URDF roll gives $\alpha_1 = 30° - 90° = -60°$.
2	1.25	q_2	$+\pi/2$	2	The joint-3 offset resolves into 1.25 along z_1 and 2 along x_2. At home $x_2 = x_1$, and the orthogonal joint axes give $\alpha_2 = +90°$.
3	0.75	$q_3 + \pi/2$	$+\pi/2$	0	Moving 0.75 along z_2 reaches the intersection with joint 4's axis: O_3 in Fig. 1(a), hence $a_3 = 0$. Choosing $x_3 = y_2$ gives the +90° angle offset and $\alpha_3 = +90°$.
4	2.5	q_4	$-\pi/2$	0	The wrist centre lies $1.5 + 1 = 2.5$ along z_3 from O_3. Axes 4 and 5 intersect there; with $x_4 = x_3$ at home, $\alpha_4 = -90°$.
5	0	q_5	$+\pi/2$	0	Axes 5 and 6 meet at the wrist centre, so $O_5 = O_4$ and $d_5 = a_5 = 0$ [Fig. 1(b)]. Keeping $x_5 = x_4$ gives $\alpha_5 = +90°$.
6	1.5	q_6	0	0	The endpoint is 1.5 along z_5 from the wrist centre. Choosing $z_6 = z_5$ and $x_6 = x_5$ at home gives $a_6 = \alpha_6 = 0$.

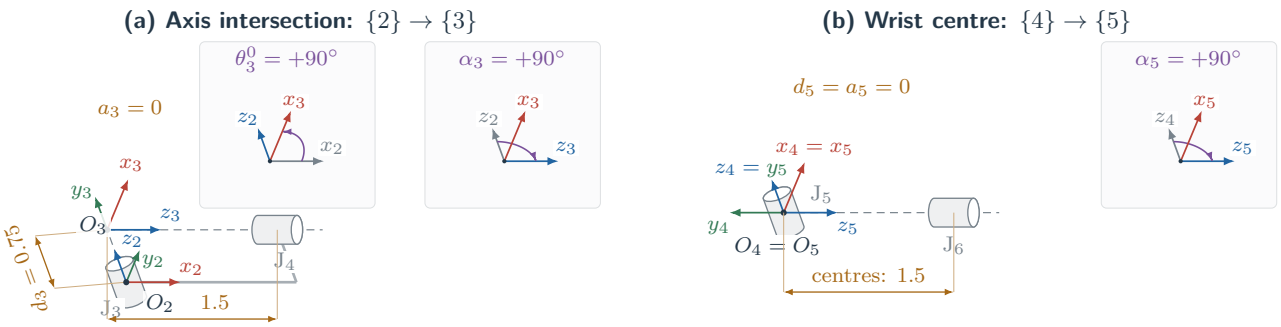


Figure 1: Frame placement at $q = 0$. (a) O_3 lies 1.5 m before the physical origin of joint 4 along its axis; this distance contributes to d_4. (b) $O_4 = O_5$ at the wrist centre; the 1.5 m between joints 5 and 6 contributes to d_6, while $d_5 = 0$. Cylinders mark physical joint origins and black dots mark DH origins; x, y, and z identify the frame axes.

Endpoint alignment. The endpoint E is `link_6`. Its origin coincides with O_6, while $x_E = z_6$, $y_E = x_6$ and $z_E = y_6$. The fixed alignment is therefore appended to the DH chain to recover the URDF endpoint pose:

$$^0T_E = \left(\prod_{i=1}^{6} {}^{i-1}T_i\right) {}^6T_E, \qquad {}^6T_E = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{2}$$

2 Analytical inverse kinematics

The axes of joints 4–6 intersect at the wrist centre, allowing position and orientation to be solved separately. We first determine the arm angles (q_1, q_2, q_3) from the desired wrist-centre position, then recover the wrist angles (q_4, q_5, q_6) from the desired orientation.

2.1 Wrist-centre positioning

Let R_d and p_d denote the desired orientation and position of `link_6` relative to `base_link`. Since the endpoint offset lies along the tool's x axis, the target wrist centre is

$$p_w = p_d - d_6 R_d e_x = (x_w, y_w, z_w)^T, \qquad e_x = (1, 0, 0)^T. \tag{3}$$

Throughout, $c_i = \cos q_i$ and $s_i = \sin q_i$.

Derivation

1. Relate the target wrist position to the three arm angles. For a candidate q_1, undo the base rotation, translate to the shoulder at $s = (a_1, 0, d_1)^T$, and undo the fixed tilt $\gamma = \pi/6$. The resulting auxiliary frame follows the base rotation and fixed tilt, but not the joint-2 rotation. With $h = z_w - d_1$ and $(c_\gamma, s_\gamma) = (\sqrt{3}/2, 1/2)$,

$$\begin{aligned} r(q_1) &:= R_x(-\gamma)\big(R_z(-q_1)p_w - s\big) = (X, Y, Z)^T, \\ X &= c_1 x_w + s_1 y_w - a_1, \\ Y &= c_\gamma(-s_1 x_w + c_1 y_w) + s_\gamma h, \\ Z &= -s_\gamma(-s_1 x_w + c_1 y_w) + c_\gamma h. \end{aligned} \tag{4}$$

These coordinates still depend on the unknown q_1.

In the moving URDF frame of `link_2`, the shoulder-to-elbow vector is $a = (a_2, d_2, 0)^T$. The elbow-to-wrist vector is $R_z(q_3)b_0$, where $b_0 = (d_4, 0, d_3)^T$ is expressed in the URDF frame of `link_3_d3`. Applying the shoulder rotation maps their sum into the auxiliary frame:

$$r(q_1) = R_y(q_2)\big(a + R_z(q_3)b_0\big) = R_y(q_2)\begin{pmatrix} a_2 + d_4 c_3 \\ d_2 + d_4 s_3 \\ d_3 \end{pmatrix}. \tag{5}$$

The rotation $R_z(q_3)$ above is expressed using the URDF link frames, with $q_3 = 0$ at the home configuration. The DH frames use a different reference orientation: at home, their angular parameter is already $\theta_3 = \pi/2$. Consequently, $\theta_3 = q_3 + \pi/2$, whereas the rotation in this geometric derivation uses q_3 directly.

2. Express $\sin q_3$ and $\cos q_3$ as functions of q_1. Rotation about y preserves both the y component and the norm. Equation (5) therefore gives two relations independent of q_2:

$$\begin{aligned} Y &= d_2 + d_4 s_3, \\ N := X^2 + Y^2 + Z^2 &= a_2^2 + d_2^2 + d_3^2 + d_4^2 + 2a_2 d_4 c_3 + 2d_2 d_4 s_3. \end{aligned} \tag{6}$$

Setting $K = d_2^2 - a_2^2 - d_3^2 - d_4^2$ yields

$$s_3 = \frac{Y - d_2}{d_4}, \qquad c_3 = \frac{N + K - 2d_2 Y}{2a_2 d_4}. \tag{7}$$

Here $a_2 = 2$ and $d_4 = 2.5$ are nonzero. For a trial value of q_1, these expressions give the required values of $\sin q_3$ and $\cos q_3$. They correspond to a real elbow angle only if $s_3^2 + c_3^2 = 1$. This consistency condition provides the equation used to determine q_1.

3. Determine the candidate base angles q_1. Since $R_x(-\gamma)$ preserves the norm, $N = \|r(q_1)\|^2 = \|R_z(-q_1)p_w - s\|^2$. Defining $N_0 = x_w^2 + y_w^2 + h^2 + a_1^2$ gives the compact expression

$$N = N_0 - 2a_1(x_w c_1 + y_w s_1). \tag{8}$$

Both N and Y are affine in (c_1, s_1). Substitution of (7) into $s_3^2 + c_3^2 = 1$ gives

$$4a_2^2(Y - d_2)^2 + (N + K - 2d_2 Y)^2 = 4a_2^2 d_4^2. \tag{9}$$

Introduce the half-angle variable $t = \tan(q_1/2)$, so that

$$D = 1 + t^2, \qquad c_1 = \frac{1 - t^2}{D}, \qquad s_1 = \frac{2t}{D}. \tag{10}$$

Writing $Y = \widetilde{Y}/D$ and $N = \widetilde{N}/D$, their quadratic numerators are

$$\begin{aligned} \widetilde{Y}(t) &= c_\gamma\big[y_w(1 - t^2) - 2x_w t\big] + \tfrac{1}{2}hD, \\ \widetilde{N}(t) &= N_0 D - 2a_1\big[x_w(1 - t^2) + 2y_w t\big]. \end{aligned} \tag{11}$$

Multiplying (9) by D^2 gives

$$\boxed{P(t) = 4a_2^2(\widetilde{Y} - d_2 D)^2 + (\widetilde{N} + KD - 2d_2\widetilde{Y})^2 - 4a_2^2 d_4^2 D^2 = 0.} \tag{12}$$

Each quantity being squared is quadratic in t, so P has degree at most four. Every real root supplies a candidate $q_1 = 2\arctan t$. The substitution $t = \tan(q_1/2)$ is undefined at $q_1 = \pi$, so this angle is not represented by any finite root of $P(t)$. To avoid missing a solution, we test $q_1 = \pi$ separately in (9) and retain it only if that equation is satisfied.

4. Recover q_2 and q_3 for each candidate q_1. For each candidate q_1, evaluate (4) and (7), then define $A = a_2 + d_4 c_3$. The remaining components of (5) are

$$X = c_2 A + s_2 d_3, \qquad Z = -s_2 A + c_2 d_3. \tag{13}$$

Thus

$$\begin{aligned} C_2 &= AX + d_3 Z = (A^2 + d_3^2)c_2, \\ S_2 &= d_3 X - AZ = (A^2 + d_3^2)s_2, \end{aligned} \qquad q_2 = \operatorname{atan2}(S_2, C_2). \tag{14}$$

The common factor is positive, so no division is needed before applying atan2. The elbow angle is recovered as $q_3 = \operatorname{atan2}(s_3, c_3)$. Since A uses c_3 directly, q_2 and q_3 can be computed in either order.

If leading polynomial coefficients vanish, the degree reduces. For this robot, $P \equiv 0$ occurs when the target wrist centre is a reachable point on the base rotation axis. At such a position, q_1 is arbitrary, yielding a continuous family of arm configurations.

Practical evaluation

Given (R_d, p_d), compute p_w from (3), followed by h, K and N_0, and form P using (11)–(12). For $P \not\equiv 0$, collect $q_1 = 2\arctan t$ from every real root and test $q_1 = \pi$ separately in (9); for $P \equiv 0$, choose any q_1, for example 0, to represent the continuous family. For each candidate, evaluate (X, Y, Z) from (4), set $N = X^2 + Y^2 + Z^2$, obtain (s_3, c_3) from (7), and set $A = a_2 + d_4 c_3$. Then compute

$$q_2 = \operatorname{atan2}(d_3 X - AZ,\ AX + d_3 Z), \qquad q_3 = \operatorname{atan2}(s_3, c_3).$$

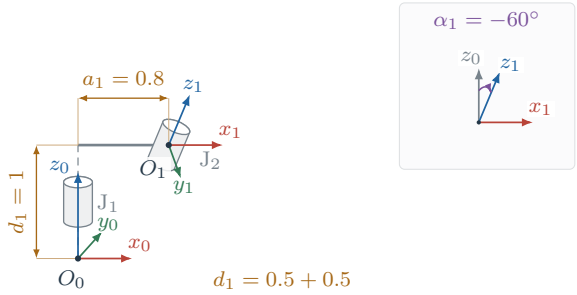
Check each triple against the target wrist position using forward kinematics, represent angles in $[-\pi, \pi]$ and remove duplicates modulo 2π. Pass the valid arm configurations to the wrist-orientation problem.

A DH frame constructions

Figure 2 collects all six frame-to-frame constructions at the home configuration ($q = 0$), using one fixed viewing direction and length scale. Lengths are in metres; angles are shown in degrees for readability.

1 Base to shoulder: $\{0\} \to \{1\}$

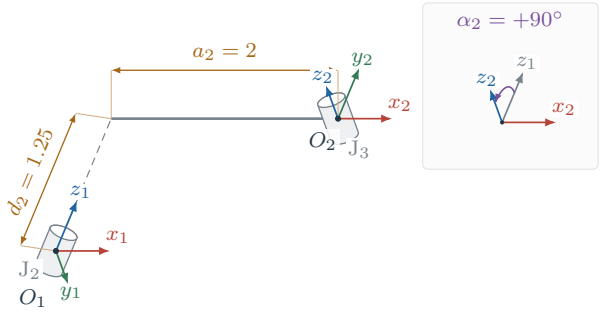
$d_1 = 1, \quad \theta_1 = q_1, \quad \alpha_1 = -60°, \quad a_1 = 0.8$



The full base height is 1 m. The 30° URDF roll of joint 2 gives $\alpha_1 = 30° - 90° = -60°$.

2 Shoulder to elbow: $\{1\} \to \{2\}$

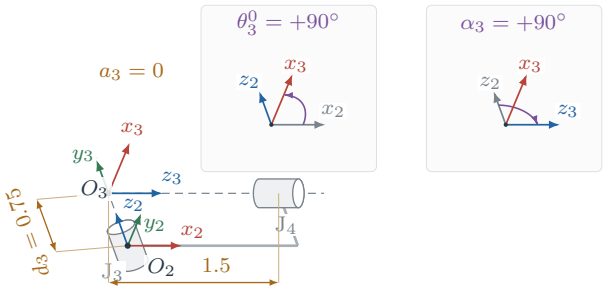
$d_2 = 1.25, \quad \theta_2 = q_2, \quad \alpha_2 = 90°, \quad a_2 = 2$



The x axes remain parallel. The offset along z_1 and the common normal along x_2 locate O_2 on joint 3.

3 Axis intersection: $\{2\} \to \{3\}$

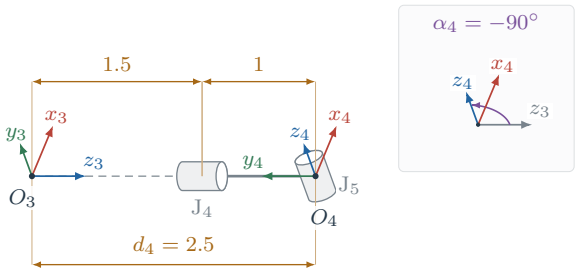
$d_3 = 0.75, \quad \theta_3 = q_3 + 90°, \quad \alpha_3 = 90°, \quad a_3 = 0$



O_3 is the intersection of the extended axes. Choosing $x_3 = y_2$ explains the fixed +90° offset in θ_3.

4 Reach the wrist: $\{3\} \to \{4\}$

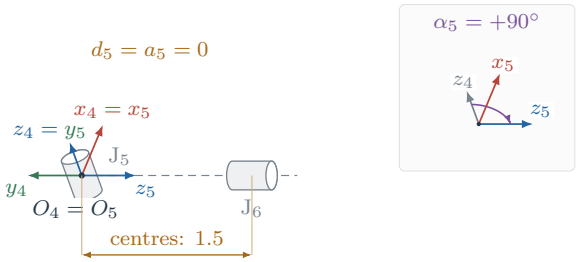
$d_4 = 2.5, \quad \theta_4 = q_4, \quad \alpha_4 = -90°, \quad a_4 = 0$



The distance starts at O_3, rather than the joint-4 cylinder. The two physical offsets add: $d_4 = 1.5 + 1$.

5 Coincident origins: $\{4\} \to \{5\}$

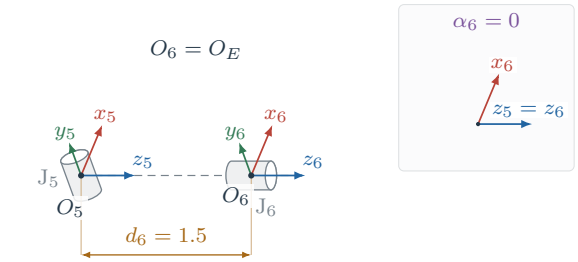
$d_5 = 0, \quad \theta_5 = q_5, \quad \alpha_5 = 90°, \quad a_5 = 0$



The origins coincide, although the joint centres do not. Here $x_4 = x_5$ and the extended joint-6 axis passes through the wrist centre.

6 Wrist to endpoint: $\{5\} \to \{6\}$

$d_6 = 1.5, \quad \theta_6 = q_6, \quad \alpha_6 = 0, \quad a_6 = 0$



At home the two frames have parallel axes. $O_6 = O_E$; the fixed endpoint alignment changes orientation only.

Figure 2: Standard DH frame assignment. Cylinders denote physical revolute joints J_i; dots mark DH origins O_i. x, y and z axes retain their colours. Dashed lines extend joint axes; gold dimensions locate the origins. Boxed angle sketches show the rotation $x_2 \to x_3$ about $+z_2$, giving θ_3^0, and the signed twist $z_{i-1} \to z_i$ about $+x_i$ (with the first direction translated to the same point). Cylinder orientation, size and viewing direction are reused between panels; drawings are schematic.